

Late Breaking Results: Advanced PCB Placement with Irregular Components for Efficient Collision Detection and Routability Optimization

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Abstract—This paper introduces an automated placement framework to optimize component positioning on modern printed circuit boards (PCBs), addressing challenges posed by heterogeneous components, irregular geometries, and complex design rules. The framework employs three key techniques to enhance placement quality and efficiency: (1) a global placement approach integrating collision detection via the Separating Axis Theorem to handle exact component contours and board shapes, (2) a multi-stage force-directed method that dynamically adjusts attractive and repulsive forces to meet clearance and routability constraints, and (3) a scanline-based legalization technique to resolve overlaps and enforce spacing requirements. Our methodology effectively adapts to diverse design limitations and accelerates the process while preserving placement quality. Experimental results demonstrate that our placer significantly improves routability over state-of-the-art solutions, demonstrating robust performance on industrial PCB designs with complex and irregular constraints.

I. INTRODUCTION

Modern PCB designs are increasingly complex due to demands for wearable and high-performance computing devices. Challenges include heterogeneous components, irregular board geometries, and intricate design rules. Manual placement is inefficient and error-prone, making automation essential for optimizing signal integrity, heat dissipation, and clearance rules.

PCB placement must ensure routability while handling irregular shapes and varying component contours. Figure 1 shows a PCB with power components on the front (Figure 1(a)) and SMDs on the back (Figure 1(b)). Through holes create vertical routing paths but impose clearance constraints. Existing placement methods can be classified into IC and PCB approaches. IC placement, using techniques like NTUplace3 [1] and ePlace-MS [2], focuses on wirelength and congestion but struggles with PCB irregularities. PCB-specific methods [3], [4] improve routability but rely on bounding box-based collision detection, which overestimates spacing, leading to inefficient layouts.

Figure 2 shows that bounding box collision detection causes excessive separation between components A and B, while exact contour modeling minimizes displacement while maintaining spacing constraints. To address these limitations, we propose a framework that accurately models component contours and resolves collisions dynamically.

Our contributions include (1) integrating collision detection using the Separating Axis Theorem (SAT) [5] with a multi-stage force-directed placement method, (2) decomposing irregular components into convex polygons to improve collision handling, (3) efficient scanline-based legalization with $O(n \log n)$ complexity to reduce layout disruptions, and (4) achieving a 38% improvement in routability compared to existing methods. Our algorithm successfully handles complex designs that prior methods could not accommodate within board constraints, demonstrating its effectiveness for modern PCB layouts.

II. PRELIMINARIES

A. Separating Axis Theorem

SAT is used for precise collision detection, determining the minimal translation vector (MTV) needed to separate overlapping objects. Unlike bounding boxes, SAT accurately captures component contours, essential in dense PCB layouts. Integrated into a multi-stage force-directed framework, SAT iteratively refines positions, resolving overlaps while optimizing board area.

B. Problem Formulation

Our PCB placement problem is formally defined as follows:

- **The Irregular-Shape-Aware PCB Placement Problem:** Given design rules, a board shape, a set of I/O components with diverse shapes,

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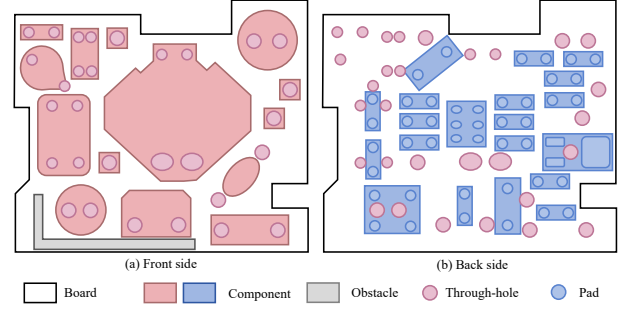


Fig. 1. (a) PCB front side with power components. (b) Back side with SMD components. Through holes (pink) connect layers but impose clearance constraints.

component pads, a clearance table for pads and through holes, and a set of netlists, place all components with optimized routability, total wirelength, and clearance violations while satisfying non-overlapping and boundary constraints.

III. ALGORITHMS

Our PCB placement flow consists of three stages: (1) *Preprocessing*, (2) *Multi-Stage Force Activation in Force-Directed Placement*, and (3) *Scanline-Based Legalization*, as shown in Figure 3.

A. Preprocessing

We define boundary constraints using *boundary extension obstacles* to handle irregular PCB shapes. This involves expanding the board outline and creating artificial obstacles at the extended boundary to restrict component placement. These obstacles ensure that components remain within the intended layout area, preserving clearance near edges and preventing unintended encroachments. Collision detection is simplified by decomposing concave polygons into convex polygons via *Ear Clipping* method [6]. Initial placement of power components follows a BFS-based approach, ensuring structured positioning.

B. Multi-Stage Force Activation in Force-Directed Placement

Our placement strategy employs a multi-stage force-directed approach that dynamically adjusts forces to optimize alignment, resolve overlaps, and enforce clearance constraints. The methodology integrates the SAT for collision detection and the MTV for resolving overlaps efficiently.

- **Stage 1:** Net attraction forces (F_{n_i}) guide interconnected components towards optimal alignment while component (F_{c_i}) and obstacle repulsion forces (F_{o_i}) ensure adequate spacing. The net attraction force for component c_i is given by:

$$F_{n_j}^x(c_i) = \alpha \cdot \frac{A_{n_j} - A_{c_i}}{A_{n_j}} \cdot (\hat{x}_{n_j} - x_{c_i}), \quad (1)$$

where A_{n_j} represents the total area of components connected by net n_j , A_{c_i} denotes the area of component c_i , and α modulates force intensity dynamically. Repulsive forces are computed using the MTV:

$$F_{\chi_i} = \beta \cdot \vec{f}_{MTV_i}, \quad \text{where } \chi \in \{c, o\}. \quad (2)$$

Overlaps are detected using the SAT, which verifies non-intersecting projections along potential separating axes. The separating axes a are derived from the edge normals of the component contours:

$$\forall \text{ axis } a, \text{ Proj}(c_i, a) \cap \text{Proj}(c_j, a) = \emptyset \implies \text{no collision}. \quad (3)$$

If projections along all separating axes indicate an intersection, the MTV is computed as:

$$\vec{v}_{MTV} = \min_{\text{axis } a} |\text{Proj}(c_i, a) \cap \text{Proj}(c_j, a)|. \quad (4)$$

TABLE I
BENCHMARK STATISTICS AND EXPERIMENTAL RESULTS

Benchmark	#Routing Layer	#Components	#Pads	#Nets	#Through Holes	TsouPCBplacer		Ours w/o MTV		Ours	
						Routability	Runtime (s)	Routability	Runtime (s)	Routability	Runtime (s)
PCB1	1	60	131	36	33	51.6%	36.6	47.4%	124.5	67.4%	49.2
PCB2	1	92	206	53	31	39.1%	52.5	23.8%	35.5	66.9%	125.6
PCB3	1	93	203	50	27	RF	63.3	37.6%	186.4	65.1%	209.4
PCB4	2	206	547	135	50	38.7%	84.9	30.5%	11.3	62.8%	90.3
PCB5	2	212	458	108	111	RF	181.1	28.7%	80.9	59.5%	186.4
PCB6	2	207	588	128	55	52.6%	67.7	40.2%	72.3	70.4%	95.3
PCB7	2	256	559	125	129	56.9%	285.1	45.3%	84.9	68.2%	89.2
Average	-	-	-	-	-	47.78%	110.3	36.21%	85.1	65.76%	120.8
Comp.	-	-	-	-	-	1.00	1.00	0.76	0.77	1.38	1.10

Note: Routing failure (**RF**) occurs when component placement exceeds the board boundaries, making the design unroutable.

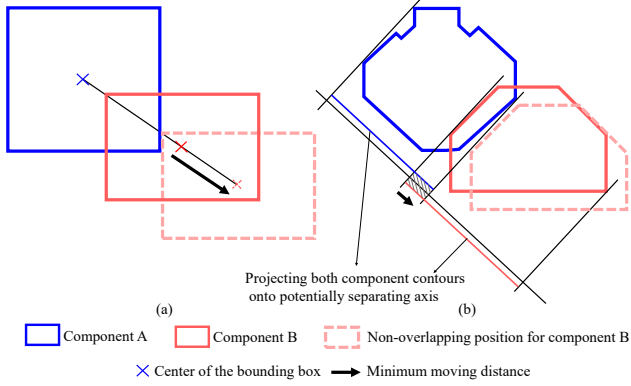


Fig. 2. (a) Bounding box collision method [4], causing wasted space. (b) SAT-based resolution, optimizing space while maintaining clearance.

This ensures accurate collision detection while minimizing displacement.

- **Stage 2:** Pad-level clearance forces (F_{p_i}) are applied to enforce design rule compliance. To integrate clearance constraints, each pad is virtually expanded by $d(p_i, p_j)/2$, ensuring sufficient separation.
- **Stage 3:** Through-hole clearance forces (F_{t_i}) are applied to maintain the required spacing between pads and through-holes. Similar to Stage 2, spacing is enforced by virtually expanding each pad and through-hole by $d(t_i, p_j)/2$.

To ensure a smooth transition in force activation, α follows a reverse sigmoid function to gradually reduce net attraction influence, while β follows a forward sigmoid function to intensify repulsion forces as placement density increases. This adaptive force tuning balances alignment and spacing dynamically, optimizing placement quality across all stages.

C. Scanline-Based Legalization

A scanline-based approach resolves remaining overlaps and spacing violations:

- Components are sorted and adjusted along the x -axes and y -axes to eliminate conflicts.
- Overlaps are detected using bounding boxes, with minimum shifts applied to enforce spacing.
- The legalization process iterates horizontally and vertically to ensure compliance.

This method guarantees manufacturable placements with minimal displacement.

IV. EXPERIMENTAL RESULTS

We implemented our placement algorithm in C++ and evaluated it on an AMD Ryzen Threadripper 3990X (4.38 GHz) Linux workstation with 125.5 GB memory. Real-world PCB benchmarks, including single-layer (PCB1-3) and double-layer (PCB4-7) designs, were used to compare our method against TsouPCBplacer [4].

Our algorithm has a theoretical complexity of $O(n^2)$ due to pairwise collision checks, where n is the number of components. However, the log-log regression analysis of component counts vs. runtimes shows an empirical complexity of only $O(n^{0.17})$. This scalability is achieved by bounding-box pre-filtering to reduce unnecessary SA Theorem computations, applying the SA Theorem only for convex polygons with bounding-box overlaps, and performing $O(n \log n)$ scanline-based legalization to ensure efficient overlap handling.

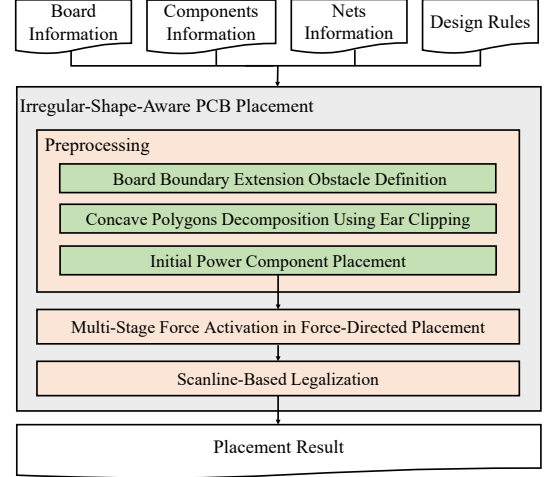


Fig. 3. Flow of our proposed placement algorithm.

Table I presents a comparison of the routability between our proposed algorithm and TsouPCBplacer [4]. The routability of ePlace [2], the state-of-the-art method, is only 87.96% of that of TsouPCBplacer. On average, our method achieves 38% higher routability compared to TsouPCBplacer.

Our method significantly improves routability, particularly in single-layer PCBs where congestion is critical. PCB2 increases from 39.1% to 66.9%, and PCB3, previously unroutable, achieves 65.1%. For double-layer PCBs, PCB4 reaches 62.8% compared to 38.7% in TsouPCBplacer, demonstrating better handling of irregular board shapes. Overall, routability improves from 47.78% to 65.76%, with only a slight runtime increase.

To evaluate MTV's effectiveness, we tested other displacement strategies on benchmark datasets, with our method as the baseline. Our approach achieved an average 81.5% improvement in routability compared to methods without MTV, computed as $(1.38 - 0.76)/0.76 = 81.5\%$. Here, 1.38 is the normalized routability improvement of our method with MTV, and 0.76 represents the routability improvement when MTV is not used (Table I). This demonstrates MTV's critical role in better component placement and routability.

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